ASSIGNMENT 1

CREATE A TABLE CALLED EMPLOYEE AND EXECUTE THE FOLLOWING. EMPLOYEE (EMPNO, ENAME, JOB, MANAGER NUMBER , SAL, COMMISSION).

```
CREATE TABLE EMPLOYEE
(EMPNO INTEGER,
 ENAME VARCHAR(30),
 JOB VARCHAR(30),
 MANAGERNUMBER INTEGER,
 SAL INTEGER,
 COMMISSION INTEGER)
```

Results Explain Describe Saved SQL History

```
Table created.
```

0.02 seconds

DESC EMPLOYEE

Results Explain Describe Saved SQL History

Object Type TABLE Object EMPLOYEE

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
EMPLOYEE	EMPNO	NUMBER	22	-	0	-	✓	-	-
	ENAME	VARCHAR2	30	-	-	-	✓	-	-
	JOB	VARCHAR2	30	-	-	-	✓	-	-
	MANAGERNUMBER	NUMBER	22	-	0	-	✓	-	-
	SAL	NUMBER	22	-	0	-	✓	-	-
	COMMISSION	NUMBER	22	-	0	-	✓	-	-

1 - 6

A) INSERT ANY THREE RECORDS IN EMPLOYEE TABLE CONTAINING ATTRIBUTES EMPNO, ENAME, JOB, MANAGER NUMBER, SAL, COMMISSION.

SQL>INSERT INTO EMPLOYEE(EMPNO, ENAME, JOB, SAL, COMMISSION) VALUES (1,'JOHN','MANAGER',1000000,10000);

1 ROW CREATED.

SQL> INSERT INTO EMPLOYEE VALUES(2,'RAMESH','SALESMAN',1,100000,1000);

1 ROW CREATED.

SQL> INSERT INTO EMPLOYEE VALUES(3,'RAJ','CLEANER',1,500000,100);

1 ROW CREATED.

SQL> SELECT * FROM EMPLOYEE;

EMPNO	ENAME	JOB	MANAGERNUMBER	SAL	COMMISSION
1	JOHN	MANAGER		1000000	10000
2	RAMESH	SALESMAN	1	100000	1000
3	RAJ	CLEANER	1	500000	100

SQL> ROLLBACK;

ROLLBACK COMPLETE.

SQL> SELECT * FROM EMPLOYEE;

NO ROWS SELECTED

B) ADD PRIMARY KEY CONSTRAINT AND NOT NULL CONSTRAINT TO THE EMPLOYEE TABLE.

ALTER TABLE EMPLOYEE MODIFY EMPNO INTEGER PRIMARY KEY

Results Explain Describe Saved SQL History

Table altered.

0.05 seconds

ALTER TABLE EMPLOYEE MODIFY ENAME VARCHAR(30) NOT NULL

Results Explain Describe Saved SQL History

```
Table altered.
```

0.00 seconds

DESC EMPLOYEE

Results Explain **Describe** Saved SQL History

Object Type **TABLE** Object **EMPLOYEE**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
EMPLOYEE	EMPNO	NUMBER	22	-	0	1	-	-	-
	ENAME	VARCHAR2	30	-	-	-	-	-	-
	JOB	VARCHAR2	30	-	-	-	✓	-	-
	MANAGERNUMBER	NUMBER	22	-	0	-	✓	-	-
	SAL	NUMBER	22	-	0	-	✓	-	-
	COMMISSION	NUMBER	22	-	0	-	✓	-	-

1 - 6

C) INSERT NULL VALUES AND PRIMARY KEY TO THE EMPLOYEE TABLE AND VERIFY THE RESULT.

INSERT INTO EMPLOYEE (EMPNO, ENAME, JOB, SAL, COMMISSION) VALUES(1, 'JOHN', 'MANAGER',1000000,10000)

Results Explain Describe Saved SQL History

```
1 row(s) inserted.
```

0.01 seconds

INSERT INTO EMPLOYEE (EMPNO, ENAME, JOB, MANAGERNUMBER, SAL, COMMISSION) VALUES(1,'RAMESH','WORKER',1,100000,1000)

Results Explain Describe Saved SQL History

```
ORA-00001: unique constraint (2BA23CS032.SYS_C007389) violated
```

.00 seconds

INSERT INTO EMPLOYEE (EMPNO, JOB, MANAGERNUMBER, SAL, COMMISSION) VALUES(1,'WORKER',1,100000,1000)

Results Explain Describe Saved SQL History

```
ORA-01400: cannot insert NULL into ("2BA23CS032"."EMPLOYEE"."ENAME")
```

0.00 seconds

ASSIGNMENT-2

CRAETE A TABLE CALLED EMPLOYEE THAT CONTAINS ATTRIBUTES EMPNO, ENAME, JOB, MANAGER SSN, SALARY AND EXECUTE FOLLOWING

CREATE TABLE NEWEMPLOYEE
(EMPNO INTEGER,
ENAME VARCHAR(20),
JOB VARCHAR(20),
MGR INTEGER,
SAL DECIMAL(8,2))

DESC NEWEMPLOYEE

Results Explain Describe Saved SQL History

Object Type TABLE Object NEWEMPLOYEE

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
NEWEMPLOYEE	EMPNO	NUMBER	22	-	0	-	✓	-	-
	ENAME	VARCHAR2	20	-	-	-	✓	-	-
	JOB	VARCHAR2	20	-	-	-	✓	-	-
	MGR	NUMBER	22	-	0	-	✓	-	-
	SAL	NUMBER	-	8	2	-	✓	-	-
									1 - 5

1.ADD A COLUMN COMMISSION WITH DOMAIN TO EMPLOYEE TABLE

ALTER TABLE NEWEMPLOYEE ADD COMM DECIMAL(5,2)

Results Explain Describe Saved SQL History

```
Table altered.
```

0.01 seconds

DESC NEWEMPLOYEE

Results Explain Describe Saved SQL History

Object Type TABLE Object NEWEMPLOYEE

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
NEWEMPLOYEE	EMPNO	NUMBER	22	-	0	-	✓	-	-
	ENAME	VARCHAR2	20	-	-	-	✓	-	-
	JOB	VARCHAR2	20	-	-	-	✓	-	-
	MGR	NUMBER	22	-	0	-	✓	-	-
	SAL	NUMBER	-	8	2	-	✓	-	-
	COMM	NUMBER	-	5	2	-	✓	-	-

1 - 6

2.INSERT ANY 5 RECORDS TO THE TABLE

INSERT INTO NEWEMPLOYEE VALUES(101,'ARUN','CLEANER',1,10000.23,123.23)

SELECT * FROM NEWEMPLOYEE

Results Explain Describe Saved SQL History

EMPNO	ENAME	JOB	MGR	SAL	COMM
101	ARUN	CLEANER	1	10000.23	123.23

1 rows returned in 0.01 seconds Download

INSERT INTO NEWEMPLOYEE VALUES(102,'ADI','ANALYST',2,10000.23,123.23)

SELECT * FROM NEWEMPLOYEE

Results Explain Describe Saved SQL History

EMPNO	ENAME	JOB	MGR	SAL	COMM
101	ARUN	CLEANER	1	10000.23	123.23
102	ADI	ANALYST	2	10000.23	123.23

2 rows returned in 0.00 seconds Download

INSERT INTO NEWEMPLOYEE VALUES(103,'ANUP','CASHIER',1,10000.23,123.23)

SELECT * FROM NEWEMPLOYEE

Results Explain Describe Saved SQL History

EMPNO	ENAME	JOB	MGR	SAL	COMM
101	ARUN	CLEANER	1	10000.23	123.23
102	ADI	ANALYST	2	10000.23	123.23
103	ANUP	CASHIER	1	10000.23	123.23

3 rows returned in 0.00 seconds Download

INSERT INTO NEWEMPLOYEE VALUES(105,'AKOP','CASHIER',3,10000.23,123.23)

SELECT * FROM NEWEMPLOYEE

Results Explain Describe Saved SQL History

EMPNO	ENAME	JOB	MGR	SAL	COMM
101	ARUN	CLEANER	1	10000.23	123.23
102	ADI	ANALYST	2	10000.23	123.23
103	ANUP	CASHIER	1	10000.23	123.23
105	AKOP	CASHIER	3	10000.23	123.23

4 rows returned in 0.00 seconds Download

INSERT INTO NEWEMPLOYEE VALUES(106,'JUDE','ATTENDER',2,10000.23,123.23)

SELECT * FROM NEWEMPLOYEE

Results Explain Describe Saved SQL History

EMPNO	ENAME	JOB	MGR	SAL	COMM
101	ARUN	CLEANER	1	10000.23	123.23
102	ADI	ANALYST	2	10000.23	123.23
103	ANUP	CASHIER	1	10000.23	123.23
105	AKOP	CASHIER	3	10000.23	123.23
106	JUDE	ATTENDER	2	10000.23	123.23

5 rows returned in 0.00 seconds Download

3.UDATE THE COLUMN DETAILS OF JOB

UPDATE NEWEMPLOYEE SET JOB='ANALYST' WHERE ENAME='ANUP'

Results Explain Describe Saved SQL History

```
1 row(s) updated.
```

0.00 seconds

SELECT * FROM NEWEMPLOYEE

Results Explain Describe Saved SQL History

EMPNO	ENAME	JOB	MGR	SAL	COMM
101	ARUN	CLEANER	1	10000.23	123.23
102	ADI	ANALYST	2	10000.23	123.23
103	ANUP	ANALYST	1	10000.23	123.23
105	AKOP	CASHIER	3	10000.23	123.23
106	JUDE	ATTENDER	2	10000.23	123.23

5 rows returned in 0.00 seconds Download

4.RENAME THE COLUMN OF EMPLOYEE TABLE

ALTER TABLE NEWEMPLOYEE RENAME COLUMN ENAME TO EMPNAME

Results Explain Describe Saved SQL History

```
Table altered.
```

0.02 seconds

DESC NEWEMPLOYEE

Results Explain Describe Saved SQL History

Object Type TABLE Object NEWEMPLOYEE

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
NEWEMPLOYEE	EMPNO	NUMBER	22	-	0	-	✓	-	-
	EMPNAME	VARCHAR2	20	-	-	-	✓	-	-
	JOB	VARCHAR2	20	-	-	-	✓	-	-
	MGR	NUMBER	22	-	0	-	✓	-	-
	SAL	NUMBER	-	8	2	-	✓	-	-
	COMM	NUMBER	-	5	2	-	✓	-	-

1 - 6

5.DELETE THE EMPLOYEE WHOSE EMPNO IS 105

DELETE FROM NEWEMPLOYEE WHERE EMPNO=105

Results Explain Describe Saved SQL History

```
1 row(s) deleted.
```

0.00 seconds

SELECT * FROM NEWEMPLOYEE

Results Explain Describe Saved SQL History

EMPNO	EMPNAME	JOB	MGR	SAL	COMM
101	ARUN	CLEANER	1	10000.23	123.23
102	ADI	ANALYST	2	10000.23	123.23
103	ANUP	ANALYST	1	10000.23	123.23
106	JUDE	ATTENDER	2	10000.23	123.23

4 rows returned in 0.00 seconds Download

ASSIGNMENT-3

QUERIES USING AGGREGATE FUNCTIONS (COUNT, AVG, MIN, MAX, SUM), GROUP BY, ORDERBY. EMPLOYEE (E_ID, E_NAME, AGE, SALARY)

CREATE A EMPLOYEE TABLE CONTAINING ALL RECORDS E_ID,E_NAME,AGE,SALARY

```
CREATE TABLE EMP1
(
 E_ID INTEGER PRIMARY KEY,
 E_NAME VARCHAR(20),
 AGE INTEGER,
 SALARY DECIMAL(8,2))
```

Results Explain Describe Saved SQL History

```
Table created.
```

0.02 seconds

DESC EMP1

Results Explain Describe Saved SQL History

Object Type TABLE Object EMP1

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
EMP1	E_ID	NUMBER	22	-	0	1	-	-	-
	E_NAME	VARCHAR2	20	-	-	-	✓	-	-
	AGE	NUMBER	22	-	0	-	✓	-	-
	SALARY	NUMBER	-	8	2	-	✓	-	-

1 - 4

INSERT INTO EMP1 VALUES(101,'ARUN',35,100000.23)

INSERT INTO EMP1 VALUES(102,'ADI',22,123456.23)

INSERT INTO EMP1 VALUES(103,'AKASH',23,234567.23)

INSERT INTO EMP1 VALUES(104,'AKSHAY',43,230040.23)

INSERT INTO EMP1 VALUES(105,'ABHI',20,200000.23)

INSERT INTO EMP1 VALUES(106,'BHUMI',20,300045.23)

SELECT * FROM EMP1

Results Explain Describe Saved SQL History

E_ID	E_NAME	AGE	SALARY
101	ARUN	35	100000.23
102	ADI	22	123456.23
103	AKASH	23	234567.23
104	AKSHAY	43	230040.23
105	ABHI	20	200000.23
106	BHUMI	20	300045.23

6 rows returned in 0.01 seconds Download

1.COUNT NUMBER OF EMPLOYEE NAMES FROM EMPLOYEE TABLE

SELECT COUNT(*) FROM EMP1

Results Explain Describe Saved SQL History

COUNT(*)
6

1 rows returned in 0.00 seconds Download

2.FIND MAXIMUM AGE FROM EMPLOYEE TABLE

SELECT MAX(AGE) FROM EMP1

Results Explain Describe Saved SQL History

MAX(AGE)
43

1 rows returned in 0.00 seconds Download

3.FIND MINIMUM AGE FROM EMPLOYEE TABLE

SELECT MIN(AGE) FROM EMP1

Results Explain Describe Saved SQL History

MIN(AGE)
20

1 rows returned in 0.00 seconds Download

4.FIND SALARY OF EMPLOYEE IN ASCENDING ORDER

SELECT SALARY FROM EMP1 ORDER BY SALARY

Results Explain Describe Saved SQL History

SALARY
100000.23
123456.23
200000.23
230040.23
234567.23
300045.23

6 rows returned in 0.00 seconds Download

INSERT INTO EMP1 VALUES(107,'ARUN',35,100000.23)

5.FIND THE GROUPED SALARIES OF EMPLOYEE

SELECT SALARY, COUNT(*) FROM EMP1 GROUP BY SALARY

Results Explain Describe Saved SQL History

SALARY	COUNT(*)
100000.23	2
234567.23	1
300045.23	1
123456.23	1
200000.23	1
230040.23	1

6 rows returned in 0.00 seconds Download

ASSIGNMENT-4

QUERRING USING INNER JOIN AND SET OPERATIONS, EMPLOYEE (SSN,ENAME,JOB,MANAGER_NO,COMMISSION), DEPARTMENT(DNUM,DNAME,MGR_SSN,MGR_STARTDATE)

1.CREATE ABOVE TABLES USING PRIMARY KEY AND FOREIGN KEY CONSTAINTS

CREATE TABLE EMP2
(SSN CHAR(5) PRIMARY KEY,
ENAME VARCHAR(15),
ADDRESS VARCHAR(20),
GENDER CHAR(1),
SALARY DECIMAL(7,2));

DESC EMP2

Results Explain **Describe** Saved SQL History

Object Type **TABLE** Object **EMP2**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
EMP2	SSN	CHAR	5	-	-	1	-	-	-
	ENAME	VARCHAR2	15	-	-	-	✓	-	-
	ADDRESS	VARCHAR2	20	-	-	-	✓	-	-
	GENDER	CHAR	1	-	-	-	✓	-	-
	SALARY	NUMBER	-	7	2	-	✓	-	-

1 - 5

CREATE TABLE DEPT2
(DNUM INTEGER PRIMARY KEY,
DNAME VARCHAR(15),
MGR_SSNCHAR(5),
MGR_STARTDATE DATE,
FOREIGN KEY (MGR_SSN) REFERENCES EMP2(SSN));
DESC DEPT2

Results Explain **Describe** Saved SQL History

Object Type **TABLE** Object **DEPT2**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
DEPT2	DNUM	NUMBER	22	-	0	1	-	-	-
	DNAME	VARCHAR2	15	-	-	-	✓	-	-
	MGR_SSN	CHAR	5	-	-	-	✓	-	-
	MGR_STARTDATE	DATE	7	-	-	-	✓	-	-
									1 - 4

2.INSERT 8 RECORDS IN EMPLOYEE & 3 RECORDS IN DEPARTMENT AND CHECK FOR REFERENTIAL CONSTRAINT VIOLATION

INSERT INTO EMP2(SSN,ENAME,ADDRESS,GENDER,SALARY) VALUES ('CC001','RAMESH','XYZ','M',10000.00)

INSERT INTO EMP2(SSN,ENAME,ADDRESS,GENDER,SALARY) VALUES ('CC002','RAM','XYZ','M',10000.00)

INSERT INTO EMP2(SSN,ENAME,ADDRESS,GENDER,SALARY) VALUES ('CC003','RAJ','XYZ','M',10000.00)

INSERT INTO EMP2(SSN,ENAME,ADDRESS,GENDER,SALARY) VALUES ('CC004','ANU','XYZ','M',10000.00)

INSERT INTO EMP2(SSN,ENAME,ADDRESS,GENDER,SALARY) VALUES ('CC005','ADI','XYZ','M',10000.00)

INSERT INTO EMP2(SSN,ENAME,ADDRESS,GENDER,SALARY) VALUES ('CC006','ARUN','XYZ','M',15000.00)

INSERT INTO EMP2(SSN,ENAME,ADDRESS,GENDER,SALARY) VALUES ('CC007','ABHI','XYZ','M',10000.00)

INSERT INTO EMP2(SSN,ENAME,ADDRESS,GENDER,SALARY) VALUES ('CC008','BHUMI','XYZ','F',31000.00)

SELECT * FROM EMP2

Results Explain Describe Saved SQL History

SSN	ENAME	ADDRESS	GENDER	SALARY
CC001	RAMESH	XYZ	M	10000
CC002	RAM	XYZ	M	10000
CC003	RAJ	XYZ	M	10000
CC004	ANU	XYZ	M	10000
CC005	ADI	XYZ	M	10000
CC006	ARUN	XYZ	M	15000
CC007	ABHI	XYZ	M	10000
CC008	BHUMI	XYZ	F	31000

8 rows returned in 0.00 seconds Download

INSERT INTO EMP2(SSN,ENAME,ADDRESS,GENDER,SALARY) VALUES ('CC003','RAMESH','XYZ','M',10000.00)

Results Explain Describe Saved SQL History

```
ORA-00001: unique constraint (2BA23CS002.SYS_C007359) violated
```

0.00 seconds

INSERT INTO DEPT2 VALUES(01,'RESEARCH','CC001','01/02/1990')
INSERT INTO DEPT2 VALUES(02,'DEVELPMENT','CC007','01/02/1990')
INSERT INTO DEPT2 VALUES(03,'MANAGEMENT','CC008','01/02/1990')

Results Explain Describe Saved SQL History

DNUM	DNAME	MGR_SSN	MGR_STARTDATE
1	RESEARCH	CC001	01/02/1990
2	DEVELPMENT	CC007	01/02/1990
3	MANAGEMENT	CC008	01/02/1990

3 rows returned in 0.01 seconds Download

INSERT INTO DEPT2 VALUES(04,'MANAGEMENT','CC009','01/02/1990')

Results Explain Describe Saved SQL History

```
ORA-02291: integrity constraint (2BA23CS002.SYS_C007361) violated - parent key not found
```

0.01 seconds

3.RETRIEVE ENAME, ADDRESS, SALARY AND DNUM OF ALL EMPLOYEE WHO ARE MANAGER

SELECT ENAME, ADDRESS, SALARY, DNAME FROM EMP2 JOIN DEPT2 ON SSN=MGR_SSN

Results Explain Describe Saved SQL History

ENAME	ADDRESS	SALARY	DNAME
RAMESH	XYZ	10000	RESEARCH
ABHI	XYZ	10000	DEVELPMENT
BHUMI	XYZ	31000	MANAGEMENT

3 rows returned in 0.00 seconds Download

4.RETRIEVE SSN OF EMPLOYEE WHO ARE NOT MANAGER

SELECT SSN FROM EMP2 MINUS SELECT MGR_SSN FROM DEPT2

Results Explain Describe Saved SQL History

SSN
CC002
CC003
CC004
CC005
CC006

5 rows returned in 0.00 seconds Download

5.RETRIEVE ENAME, SALARY, DNMAE OF MANAGERS WHOSE SALARY IS GREATER THAN 30000

SELECT ENAME, SALARY, DNAME FROM EMP2 JOIN DEPT2 ON SSN=MGR_SSN AND SALARY >30000.00

Results Explain Describe Saved SQL History

ENAME	SALARY	DNAME
BHUMI	31000	MANAGEMENT

1 rows returned in 0.00 seconds Download

6.RETRIEVE ALL ENAME AND DNAME FROM BOTH THE TABLE

SELECT ENAME FROM EMP2 UNION SELECT DNAME FROM DEPT2

Results Explain Describe Saved SQL History

ENAME
ABHI
ADI
ANU
ARUN
BHUMI
DEVELPMENT
MANAGEMENT
RAJ
RAM
RAMESH
RESEARCH

11 rows returned in 0.00 seconds Download

ASSIGNMENT-5

CREATE A ROW LEVEL TRIGGER FOR CUSTOMER TABLE THAT WOULD FIRE FOR INSERT/UPDATE/DELETE OPERATIONS PERFORMED ON THE CUSTOMERS TABLE. THIS TRIGGER WILL DISPLAY SALARY DIFFERENCE BETWEEN OLD AND NEW SALARY CUSTOMER(ID,NAME,AGE,ADDRESS,SALARY)

```
CREATE TABLE CUSTOMERS(
ID CHAR(5) PRIMARY KEY,
NAME VARCHAR(12),
AGE INTEGER,
ADDRESS VARCHAR(20),
SAL DECIMAL(6,2))
DESC CUSTOMERS
```

Results Explain **Describe** Saved SQL History

Object Type **TABLE** Object **CUSTOMERS**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
CUSTOMERS	ID	CHAR	5	-	-	1	-	-	-
	NAME	VARCHAR2	12	-	-	-	✓	-	-
	AGE	NUMBER	22	-	0	-	✓	-	-
	ADDRESS	VARCHAR2	20	-	-	-	✓	-	-
	SAL	NUMBER	-	6	2	-	✓	-	-
									1 - 5

```
CREATE OR REPLACE TRIGGER CUST_OP
BEFORE INSERT OR UPDATE OR DELETE ON CUSTOMERS
FOR EACH ROW
DECLARE
OS CUSTOMERS.SAL%TYPE;
NS CUSTOMERS.SAL%TYPE;
```

```
SD CUSTOMERS.SAL%TYPE;
BEGIN
IF INSERTING THEN
DBMS_OUTPUT.PUT_LINE('ROW INSERTED TO CUSTOMERS TABLE SUCCESSFULLY');
ELSIF DELETING THEN
DBMS_OUTPUT.PUT_LINE('ROW DELETED FROM CUSTOMERS TABLE SUCCESSFULLY');
ELSIF UPDATING THEN
OS:=:OLD.SAL;
NS:=:NEW.SAL;
SD:=NS-OS;
DBMS_OUTPUT.PUT_LINE('OLD SALARY IS '||OS);
DBMS_OUTPUT.PUT_LINE('NEW SALARY IS '||NS);
DBMS_OUTPUT.PUT_LINE('SALARY DIFFERENCE IS '||SD);
END IF;
END;
```

Results Explain Describe Saved SQL History

```
Trigger created.
```

0.00 seconds

```
INSERT INTO CUSTOMERS VALUES('ID001','ABHI',19,'JKD',7000.00)
```

Results Explain Describe Saved SQL History

```
ROW INSERTED TO CUSTOMERS TABLE SUCCESSFULLY

1 row(s) inserted.
```

0.00 seconds

INSERT INTO CUSTOMERS VALUES('ID002','BHUMI',19,'JKD',6000.00)

Results Explain Describe Saved SQL History

```
ROW INSERTED TO CUSTOMERS TABLE SUCCESSFULLY

1 row(s) inserted.
```

0.00 seconds

SELECT *FROM CUSTOMERS

Results Explain Describe Saved SQL History

ID	NAME	AGE	ADDRESS	SAL
ID001	ABHI	19	JKD	7000
ID002	BHUMI	19	JKD	6000

2 rows returned in 0.00 seconds Download

DELETE FROM CUSTOMERS WHERE ID='ID001'

Results Explain Describe Saved SQL History

```
ROW DELETED FROM CUSTOMERS TABLE SUCCESSFULLY

1 row(s) deleted.
```

0.00 seconds

SELECT *FROM CUSTOMERS

Results Explain Describe Saved SQL History

ID	NAME	AGE	ADDRESS	SAL
ID002	BHUMI	19	JKD	6000

1 rows returned in 0.00 seconds Download

UPDATE CUSTOMERS SET SAL=9000.00 WHERE ID='ID002'

Results Explain Describe Saved SQL History

```
OLD SALARY IS 6000
NEW SALARY IS 9000
SALARY DIFFERENCE IS 3000

1 row(s) updated.
```

0.00 seconds

ASSIGNMENT-6

CREATE A CURSOR FOR EMPLOYEE TABLE AND EXTRACT VALUES FROM THE TABLE. DECLARE THE VARIABLE, OPEN THE CURSOR AND EXTRACT THE VALUES FROM THE CURSOR, CLOSE THE CURSOR

EMPLOYEE(E_ID,E_NAME,AGE,SALARY)

```
CREATE TABLE EMPLOYEE(
E_ID CHAR(5) PRIMARY KEY,
E_NAME VARCHAR(12),
AGE INTEGER,
SAL DECIMAL(6,2))
DESC EMPLOYEE
```

Results Explain Describe Saved SQL History

Object Type **TABLE** Object **EMPLOYEE**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
EMPLOYEE	E_ID	CHAR	5	-	-	1	-	-	-
	E_NAME	VARCHAR2	12	-	-	-	✓	-	-
	AGE	NUMBER	22	-	0	-	✓	-	-
	SAL	NUMBER	-	6	2	-	✓	-	-

1 - 4

```
INSERT INTO EMPLOYEE VALUES('ID001','ABHI',20,7000.00)
INSERT INTO EMPLOYEE VALUES('ID002','BHUMI',20,8000.00)
INSERT INTO EMPLOYEE VALUES('ID003','AK',23,9000.00)
INSERT INTO EMPLOYEE VALUES('ID004','SANGU',20,5000.00)
SELECT *FROM EMPLOYEE
```

Results Explain Describe Saved SQL History

E_ID	E_NAME	AGE	SAL
ID001	ABHI	20	7000
ID002	BHUMI	20	8000
ID003	AK	23	9000
ID004	SANGU	20	5000

4 rows returned in 0.00 seconds Download

```
DECLARE
CURSOR EMP_C IS SELECT E_ID,E_NAME,AGE,SAL FROM EMPLOYEE;
EMPID EMPLOYEE.E_ID%TYPE;
EMPNAME EMPLOYEE.E_NAME%TYPE;
EMPAGE EMPLOYEE.AGE%TYPE;
EMPSAL EMPLOYEE.SAL%TYPE;
BEGIN
DBMS_OUTPUT.PUT_LINE('----------');
DBMS_OUTPUT.PUT_LINE('EMP_ID      EMP_NAME      EMP_AGE      EMP_SAL');
DBMS_OUTPUT.PUT_LINE('----------');
OPEN EMP_C;
LOOP
FETCH EMP_C INTO EMPID,EMPNAME,EMPAGE,EMPSAL;
EXIT WHEN EMP_C%NOTFOUND;
DBMS_OUTPUT.PUT_LINE(EMPID||'      '||EMPNAME||'      '||EMPAGE||'      '||EMPSAL);
END LOOP;
DBMS_OUTPUT.PUT_LINE('----------');
DBMS_OUTPUT.PUT_LINE('NUMBERS OF ROWS IN EMPLOYEE TABLE IS:'||EMP_C%ROWCOUNT);
CLOSE EMP_C;
END;
/
```

Results Explain Describe Saved SQL History

```
----------
EMP_ID      EMP_NAME     EMP_AGE     EMP_SAL
----------
ID001     ABHI     20     7000
ID002     BHUMI     20     8000
ID003     AK     23     9000
ID004     SANGU     20     5000
----------
NUMBERS OF ROWS IN EMPLOYEE TABLE IS:4

Statement processed.
```

ASSIGNMENT-7

WRITE A PL/SQL BLOCK OF CODE USING PARAMETRISED CURSOR THAT WILL MERGE THE DATA AVAILABLE IN THE NEWLY CREATED TABLE N_ROLLCALL WITH THE DATA AVAILABLE IN THE TABLE O_ROLLCALL. IF THE DATA IN THE FIRST TABLE ALREADY EXIST IN THE SECOND TABLE THEN THE DATA SHOULD BE SKIPPED.

```
CREATE TABLE O_ROLLCALL
(RN INT,
NAME VARCHAR(10),
ADDRESS VARCHAR(15))
DESC O_ROLLCALL
```

Results Explain Describe Saved SQL History

Object Type TABLE Object O_ROLLCALL

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
O_ROLLCALL	RN	NUMBER	22	-	0	-	✓	-	-
	NAME	VARCHAR2	10	-	-	-	✓	-	-
	ADDRESS	VARCHAR2	15	-	-	-	✓	-	-
									1 - 3

```
INSERT INTO O_ROLLCALL VALUES(1,'ABHI','BGK')
INSERT INTO O_ROLLCALL VALUES(2,'ADI','JKD')
SELECT *FROM O_ROLLCALL
```

Results Explain Describe Saved SQL History

RN	NAME	ADDRESS
1	ABHI	BGK
2	ADI	JKD

2 rows returned in 0.00 seconds Download

```
CREATE TABLE N_ROLLCALL
(RN INT,
NAME VARCHAR(10),
ADDRESS VARCHAR(15))
DESC N_ROLLCALL
```

Results Explain **Describe** Saved SQL History

Object Type **TABLE** Object **N_ROLLCALL**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
N_ROLLCALL	RN	NUMBER	22	-	0	-	✓	-	-
	NAME	VARCHAR2	10	-	-	-	✓	-	-
	ADDRESS	VARCHAR2	15	-	-	-	✓	-	-
									1 - 3

```
INSERT INTO N_ROLLCALL VALUES(1,'ABHI','BGK')
INSERT INTO N_ROLLCALL VALUES(3,'ARUN','JKD')
SELECT *FROM O_ROLLCALL
```

Results Explain Describe Saved SQL History

RN	NAME	ADDRESS
1	ABHI	BGK
3	ARUN	JKD

2 rows returned in 0.00 seconds Download

```
DECLARE
CURSOR O_C(R N_ROLLCALL.RN%TYPE) IS SELECT RN,NAME,ADDRESS FROM
N_ROLLCALL WHERE RN>R;
ROLL N_ROLLCALL.RN%TYPE;
R_N N_ROLLCALL.RN%TYPE;
NM N_ROLLCALL.NAME%TYPE;
ADR N_ROLLCALL.ADDRESS%TYPE;
CT INT:=0;
BEGIN
R_N:=:r;
OPEN O_C(R_N);
```

```
LOOP
FETCH O_C INTO ROLL,NM,ADR;
EXIT WHEN O_C%NOTFOUND;
SELECT COUNT(*) INTO CT FROM O_ROLLCALL WHERE RN=ROLL AND
NAME=NM AND ADDRESS=ADR;
IF CT=0 THEN
INSERT INTO O_ROLLCALL VALUES(ROLL,NM,ADR);
DBMS_OUTPUT.PUT_LINE(ROLL||' '||NM||'   '||ADR);
DBMS_OUTPUT.PUT_LINE('ROW INSERTED TO OLD ROLLCALL');
END IF;
END LOOP;
IF CT>0 THEN
DBMS_OUTPUT.PUT_LINE('ROW ALREADY PRESENT IN OLD ROLLCALL');
END IF;
CLOSE O_C;
END;
/
```

Results Explain Describe Saved SQL History

```
3 ARUN   JKD
ROW INSERTED TO OLD ROLLCALL

Statement processed.
```

0.01 seconds

SELECT *FROM O_ROLLCALL

Results Explain Describe Saved SQL History

RN	NAME	ADDRESS
1	ABHI	BGK
2	ADI	JKD
3	ARUN	JKD

3 rows returned in 0.01 seconds Download